

Workflow and Process Automation

SUBTITLE

Heading 1

Heading 2

Heading 3

Heading 4

Tip 1

SOQL Queries typically only return active records, non-archived records.

Tip 2

Dont use limit w/o order by because the record might change before u run.

Tip 3

```
select name, account.name, createddate from contact where createdDate = Today
```

```
select name, account.name, createddate from contact where createdDate =  
LAST_N_Days:60.
```

```
select name, account.name, createddate from contact where createdDate <  
LAST_N_Days:60
```

```
select name, account.name, createddate from contact where createdDate =  
LAST_N_Months:1
```

```
select name, account.name, createddate from contact where createdDate =  
LAST_N_Months:1
```

```
select name, account.name, createddate from contact where createdDate =  
LAST_FISCAL_QUARTER
```

Tip 4

Record trigger → Scheduled path is equivalent to time triggered workflow rule or time based actions in process builder. e.g renewal of membership.

Its different than scheduled triggered for which run based on a schedule.

RECAP:

- ☒ Scheduled Paths are used in Record-Triggered Flows to automate steps at a designated time.
- ☒ A record must be created or updated to set it on a Scheduled Path.
- ☒ Scheduled Paths can be based on any date or date/time field on the object.
- ☐ Test your flows in a sandbox first.

Workflow and Process Automation Introduction

	Declarative tools like workflow rules and process builder. Both of which sf is retiring in favor of flows.

Why Flow retired Workflow rules and process builder

Flow	The “Apex for Admins” tool — declarative automation that behaves like code but without coding.
Flow	Can do everything workflow rules and process builder could do and more.
	CRUD Loop Branching using decision logic. Call invocable apex. Send email. Interact with users. Run before/after save Submit for approval. Work in Screen mode and background mode.
5 types of Flow	1. Record triggered flow 2. Scheduled triggered flow 3. Screen flow 4. Autolaunched flow 5. Platform event-triggered flow
Record triggered flow	Runs before/after.
Scheduled triggered flow	Run at scheduled time (daily/weekly)
Screen flow	Shows ui screen to users. Used for guided interactions.(intake forms)

Autolaunched flow If a flow has screens → it is a Screen Flow. If a flow has NO screens → it is an Autolaunched Flow (regardless of trigger).	Runs w/o user interaction. Called by Apex, buttons, subflows, or processes. It can still be <i>triggered</i> by a user action, but it never <i>interacts</i> with the user once running.
Platform Event-Triggered flow	Runs when platform event is triggered. Used for event-driven integrations.

Mnemonic	ARSSP (Always remembers salesforce standardize properly)
	A: Autolaunched flow R: Record triggered flow S: Scheduled triggered flow S: Screen flow P: Platform event-driven flow

Jargon

Element	A step inside the flow • Assignment • Decision • CRUD • Loops • Screen						
Resource	Variables used inside the flow • Variables • Constants • Text templates • Record variables(obj.field) • Collections([obj,obj])						
Fault path	Flows error handling branch						
Subflow	A flow that is used by another flow. Used to modularize automation.						
	<table> <tr> <th>Trigger Timing</th><th>Used For</th></tr> <tr> <td>Before Save</td><td>Fast updates to the same record</td></tr> <tr> <td>After Save</td><td>Creating related records, sending emails, calling Apex</td></tr> </table>	Trigger Timing	Used For	Before Save	Fast updates to the same record	After Save	Creating related records, sending emails, calling Apex
Trigger Timing	Used For						
Before Save	Fast updates to the same record						
After Save	Creating related records, sending emails, calling Apex						

Process automation settings

Scenario: A Workflow Rule on the Opportunity object is set up to send a "Follow-up Email" 7 days before the CloseDate. The Problem: Six days later, Sarah leaves the company and her user account is deactivated. The action is scheduled to run tomorrow. The system uses the Default Workflow User to execute the action (send the email). This ensures the business process completes without interruption.	The default workflow user is required for scheduled paths in record-triggered flows and time-dependent actions in workflow rules. If the user who triggered the automation is no longer active, the default workflow user is associated with the actions that are executed by the automation. The default workflow user always runs schedule-triggered flows. A time dependent workflow action involves two stages. 1. Workflow criteria is evaluated in sarahs context and permission. System creates a time based queue item. She is in the entry of the one who trigger it. 2. if she is inactive, default workflow user with his PS executes it.
The workflow isnt triggered by action but by clock.	Schedule triggered flow always run as Default user.
When this setting is enabled, users with the Manage Flow and View . Let admins debug flows as other users ☒	When this setting is enabled, users with the Manage Flow and View All Data permissions can specify the running user each time they debug a flow.

Workflow Rules, Action and retirement

Workflow automates the following types of actions based on your organization's processes:

- ⚡ Tasks—Assign a new task to a user, role, or record owner.
- ⚡ Email Alerts—Send an email to one or more recipients you specify.
- ⚡ Field Updates—Update the value of a field on a record.
- ⚡ Outbound Messages—Send a secure, configurable API message (in XML format) to a designated listener.

For example, workflow can:

- ⚡ Assign follow-up tasks to a support rep one week after a case is updated.
- ⚡ Send sales management an email alert when a sales rep qualifies a large deal.
- ⚡ Change the Owner field on a contract three days before it expires.
- ⚡ Trigger an outbound API message to an external HR system to initiate the reimbursement process for an approved expense report.

Each workflow rule consists of:

- ⚡ Criteria that cause the workflow rule to run.
- ⚡ Immediate actions that execute when a record matches the criteria. For example, Salesforce can automatically send an email that notifies the account team when a new high-value opportunity is created.
- ⚡ Time-dependent actions that queue when a record matches the criteria, and execute according to time triggers. For example, Salesforce can automatically send an email reminder to the account team if a high-value opportunity is still open ten days before the close date.

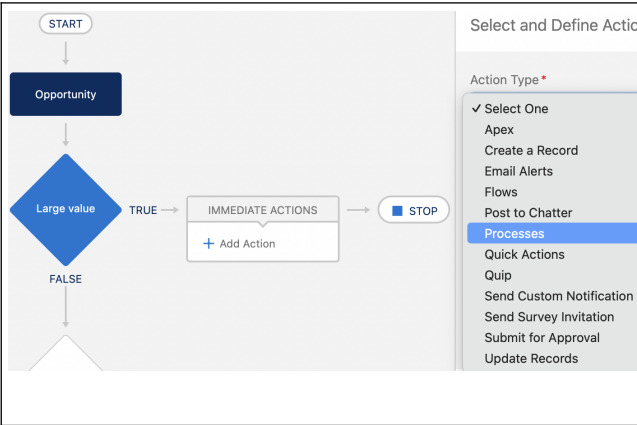
ETFO	Email Alerts Fields updates Task to user Outbound messages to external system.
Select the object to which this workflow rule applies. Object Opportunity	Select the object to which workflow rule applies.
Evaluation Criteria Evaluate the rule when a record is: ☐ created ☐ created, and every time it's edited ☒ created, and any time it's edited to subsequently meet criteria How do I choose?	1st option: Created run only when its created and meets the criteria. 3rd option: runs when its created and meets the criteria but also when the criteria field is edited from not meeting the critier to meet the criteria. So unrelated edits on the record wont trigger it.

	2nd option: created, and every time it's edited With this option, the rule repeatedly runs every time a record is edited as long as the record meets the rule criteria. If you select this option, you can't add time-dependent actions to the rule.
	3rd option: With this option, the rule can run multiple times per record, but it doesn't run when the record edits are unrelated to the rule criteria. For example, suppose that for an opportunity record to meet the rule criteria, the opportunity probability must be greater than 50%. If you create an opportunity with a probability of 75%, the workflow rule runs. If you edit that opportunity by changing the probability to 25%, the edit doesn't cause the rule to run. If you then edit that opportunity by changing the probability from 25% to 75%, the edit causes the rule to run. With this last edit, the rule runs, because the record is changed from not meeting the rule criteria to meeting the rule criteria.
Rule Criteria Run this rule if the criteria are met : Field Opportunity: Stage Operator equals Value Closed Won	Set the criteria
Specify the workflow actions that will be triggered when the rule criteria are met. See an example Rule Criteria Opportunity: Stage EQUALS Closed Won Evaluation Criteria Evaluate the rule when a record is created, and any time it's edited to subsequently meet criteria Immediate Workflow Actions No workflow actions have been added. Add Actions Time-Dependent Workflow Actions See an example No workflow actions have been added. Before adding a workflow action, you must have at least one time trigger defined. Add Time Trigger	You can either create Immediate Workflow actions Or Time-dependent workflow actions.
Immediate Workflow Actions No workflow actions have been added. Add Actions New Task New Email Alert New Field Update New Outbound Message New Flow Select Existing Action	ETFO + FLOW + existing actions.
Edit Task Object Opportunity Assigned To Kazim Syed Subject Send welcome letter Unique Name Send_welcome_letter Due Date Opportunity: Close Date plus 1 days Protected Component Status Not Started Priority Normal	The thing about task, its the only thing that a workflow rule can create an object for. Whereas process builder and flows can create new records for any object. Workflow rule for any task.

	Create a task to associate with one or more workflow rules, approval processes, or entitlement processes. When changing a task, any modifications will apply to all rules, approvals, or entitlement processes associated with it.
Now mike is doing a field update	
Field Update Edit SaveSave & NewCancel Identification NameProspect to Client Unique NameProspect_to_Client DescriptionThis changes the account record type to client ObjectOpportunity Field to UpdateAccountAccount Record Type Field Data TypeRecord Type Specify New Field Value Account Record TypeClient	Define the field update, including the object associated with the workflow rule, approval process, or entitlement process, the field to update, and the value to apply. Note that the field to update may be on a related object. Fields are shown only for the type that you select.
	Cross object field update.
EditDeleteCloneActivate Activate	Activate the workflow
RelatedDetails Account Record Type Client	It worked. I created an opportunity for ABM company and set the stage to closed won and the workflow rule set the stage to client.
RefreshExpand AllView All Upcoming & Overdue > Send welcome letter You have an upcoming task about WFR To change what's shown, try changing your filters.	We have a task that we are designated here.

Process Builder actions and retirement

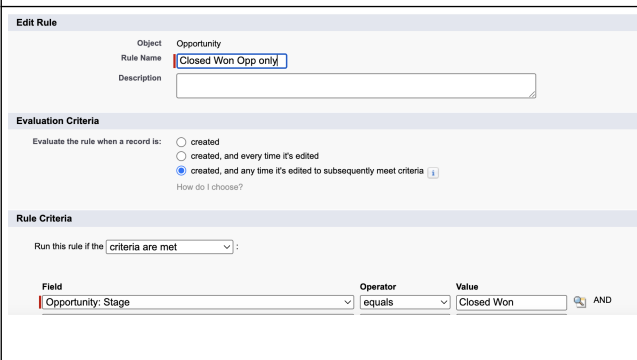
New Process Description The process starts when * ✓ Select One A record changes A platform event message is received It's invoked by another process	Mike selected record changes
Expand AllCollapse All START ↓ + Add Object ↓ + Add Criteria TRUE → IMMEDIATE ACTIONS + Add Action TIME → SCHEDULED ACTIONS ⌚ Set Schedule + Add Action → STOP FALSE → STOP	Process builder
Choose Object and Specify When to Start the Process Object * ⓘ Opportunity Start the process * ☐ only when a record is created ☒ when a record is created or edited	Select the object
☒ Conditions are met ☐ Formula evaluates to true ☐ No criteria—just execute the actions! Set Conditions Field *Operator *Type *Value * 1 [Opportunity]... QGreater thanCurrency100,000 X + Add Row Conditions * ☒ All of the conditions are met (AND) ☐ Any of the conditions are met (OR)	Opportunity.amt > 100,000
Customize the logic for the filter conditions that you specified. For example, if you enter 1 AND (2 OR 3), this criteria evaluates to true when the first condition is true and either the second or third condition is true.	Set custom logic.



He didn't finish process builder.

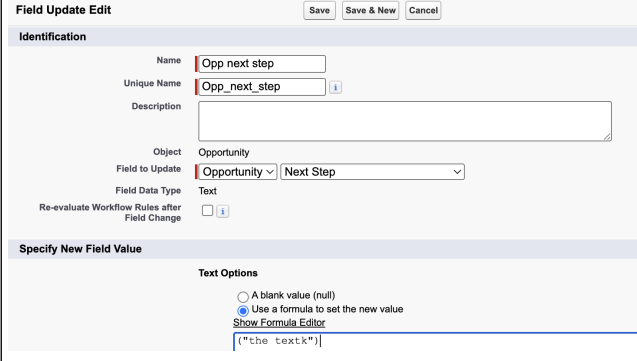
Migrate to flow

Process Automation → Migrate to flow



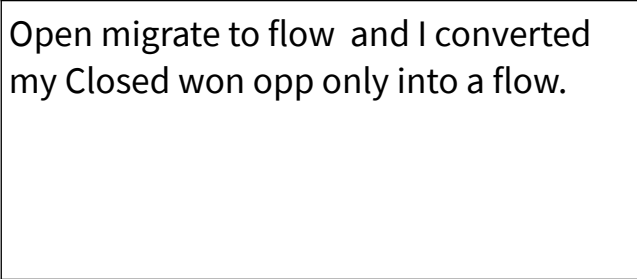
Cross object reference workflow doesn't properly migrate, so we would be modifying that workflow of ours.

Field Update Edit

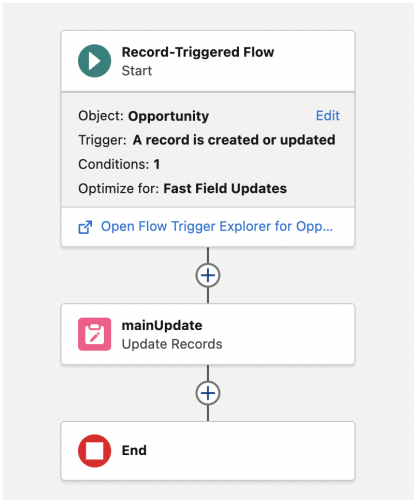


Workflow action.
I deactivated the existing flow.

Open migrate to flow and I converted my Closed won opp only into a flow.



Flow builder

Process Automation > Flow builder	
We received a record-triggered flow.	
Edit record-triggered flow	Why use Set entry level Specify entry conditions to reduce the number of records that trigger the flow and the number of times the flow is executed. Minimizing unnecessary flow executions helps to conserve your org's resources.
Configure Trigger Trigger the Flow When: ☐ A record is created ☐ A record is updated ☒ A record is created or updated ☐ A record is deleted	
When to Run the Flow for Updated Records ⓘ ☐ Every time a record is updated and meets the condition requirements ☒ Only when a record is updated to meet the condition requirements	I <3 second one.
Optimize Flow Optimize the Flow for: Fast Field Updates Update fields on the record that triggers the flow to run. This high-performance flow runs before the record is saved to the database.	Record trigger before save – used when the record that triggers the flow needs to be updated.

Actions and Related Records Update any record and perform actions, like send an email. This more flexible flow runs after the record is saved to the database.	
Now we editing update records	
* How to Find Records to Update and Set Their Values ☒ Use the opportunity record that triggered the flow ☐ Update records related to the opportunity record that triggered the flow ☐ Use the IDs and all field values from a record or record collection ☐ Specify conditions to identify records, and set fields individually	Because this flow runs before a record is saved, you can only update the record that triggered the flow to run. To update other records, configure the trigger to run the flow after the record is saved.
	You can add resources as well.
Closed Won Opp only The flow failed to start because the provided values were not valid.	The flow failed to start because the provided values were not valid: You can't launch a record-triggered flow from within Flow Builder. To launch this flow, activate it and then create, delete, or update a record, depending on the trigger configuration.
	U have to deactivate the exisiting workflow.

Specify the workflow actions that will be triggered when the rule criteria are met. [See an example](#)

Rule Criteria (Opportunity: Stage NOT EQUAL TO Closed Won) AND (Opportunity: Amount GREATER THAN 1,000,000)

Evaluation Criteria Evaluate the rule when a record is created, and any time it's edited to subsequently meet criteria

Immediate Workflow Actions

Action	Type	Description
Edit Remove	Email Alert	Notify Account Team stating that a new high value opportunity has been created
Edit Remove	Email Alert	Notify Executive Committee
Edit Remove	Field Update	Change Probability to 20%
Edit Remove	Outbound Message	Update Marketing Systems

Add Workflow Action ▼

Time-Dependent Workflow Actions [See an example](#)

 14 Days Before Opportunity: Close Date [Edit](#) | [Delete](#)

Action	Type	Description
Edit Remove	Task	Opportunity Owner should follow up with customer

Add Workflow Action ▼

 7 Days Before Opportunity: Close Date [Edit](#) | [Delete](#)

Action	Type	Description
Edit Remove	Task	Sales Manager should contact the account
Edit Remove	Field Update	Flag for executive involvement

Add Workflow Action ▼

Add Time Trigger

You can have both Immediate and time-dependent actions.

Testing our flow

On the process builder, open tests.	We can test records instead of updating records to trigger record-level trigger.
Select whether the test is triggered to run when a record is created or updated. A test can take only the flow path that runs immediately. * Run the Test When a Record Is: ☐ Created ☒ Updated Path for Test Run Immediately	
Set Test Details, Trigger, and Path Set Initial Triggering Record Set Updated Triggering Record Set Assertions	Set the initial values for the record that triggers the test. If you convert the debug run to a test, the values are set from the debug run's triggering record. After the test finishes, all transactions are rolled back, and record changes aren't saved. I selected awam foundation which isnt closed yet.
Set Updated Triggering Record Set Assertions Power supply product interest --None-- Probability (%) 100% * Stage Closed Won	
All Resources > Triggering Opportunity Record Fields Next Step {!\$Record.NextStep} Many fields comes up.	Global Variables API Prior Values of Triggering Opportunity Running Flow Interview Running Org Running User Running User Profile Running User Role System Search resources... Enter a value. Add Custom Failure Message
All Resources Global Constants False True Blank Value (Empty String) Blank Value (Empty String) Search resources...	

• Assertion: 1

• Resources

Triggering Opportunity > Next Step

Operator

Does Not Equal

• Value

Blank Value (Empty String)

Add Custom Failure Message

+ Add Assertion

To test a flow, set conditions and custom failure messages for each assertion. At run time, the test checks all of its assertions. If all conditions for each assertion evaluate to true, the assertion passes. If one condition for an assertion evaluates to false, the assertion and test run fail. If a failure occurs for an assertion during the test run, the custom failure message appears in the test run details.

Create

☒	Test L...	Descr...	Creat...	Last ...	Last R...	Result	
☒	ClosedWo...	Kazim Syed	06/12/202...	06/12/202...	✓	Pass	

Run

CancelRun

I ran the test and the result says pass.

Mike wasn't able to pass. So he used the `isChanged` operator instead of `does not equals`.

Types of flow

Get Started with Automations

Select a category, flow type, use search, or let Einstein build an automation for you.

Categories

Triggered Automations

Automations launched by records and events. This type of automation runs without user interaction.

[View All Automations](#)

Scheduled Automations

Time-based automations that launch at a specific time or frequency. This type of automation runs without user...

[View All Automations](#)

Screen Automations

Interface-driven automations that guide users through business processes. This type of automation collects or displays...

[View All Automations](#)

Autolaunched Automations

Automations that automatically launch when invoked by APIs, templates, processes, conditions, or something el...

[View All Automations](#)

Frequently Used

Record-Triggered Flow

Launches when a record is created, updated, or deleted. This autolaunched flow runs in the background.

Screen Flow

Guides users through a business process that's launched from Lightning pages, Experience Cloud sites, quick...

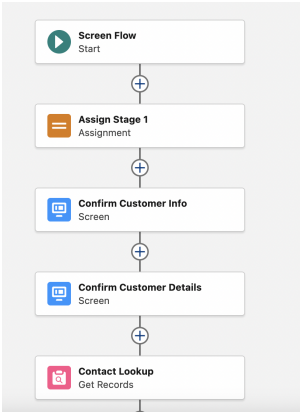
Schedule-Triggered Flow

Launches at a specified time and frequency for each record in a batch. This autolaunched flow runs in the...

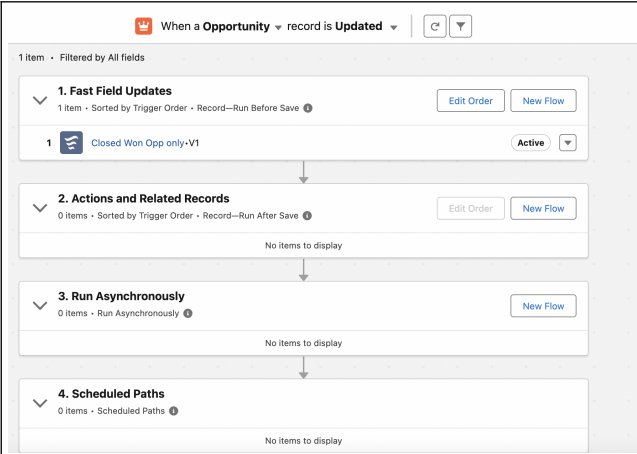
Autolaunched Flow (No Trigger)

Launches when invoked by Apex, processes, REST API, and more. This autolaunched flow runs in the...

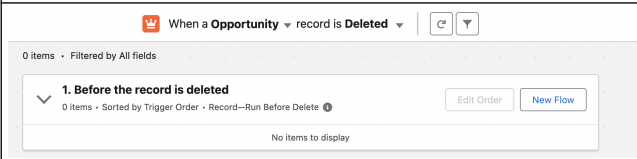
[View All Automations](#)

	Our contract corner flow.
You can goto app exchange and search flows	U will spot flows by saleslab which is from salesforce like Questionnaire flow provided by ibm. Profile wizard where u make updates to profile inside of a flow.
Omni channel flow	Where u route chates to agents with right skills and lot more around case routing, chat routing.
How to control the order of the flow in which they execute. Since we have auto launched flows that are happening behind the scene in automations.	Next lecture.

Flow trigger explorer



Set the order in which flow execute in ur org. when a record is updated.



We can change when record is updated to delete and we see we only have run-before delete.

IF YOU WANT SOMETHING TO HAPPEN: When a new case is created. When a lead's status field is changed. When a record is created or updated. When a record is deleted. USE THE TRIGGER: When a record is created. When a record is updated. When an account is created or the account priority field is changed. When a contact is deleted.	USING THIS FLOW TYPE: Record-Triggered Record-Triggered count Record-Triggered Record-Triggered
When a lead's status field is changed. When a record is created or updated. When a record is deleted. A week after a quote is created. Every Saturday at midnight. When a record is updated. When an account is created or the account priority field is changed. When a contact is deleted. After a certain amount of time. At a specified time and frequency.	Record-Triggered Record-Triggered Record-Triggered Record-Triggered. Note: Add a scheduled path Schedule-Triggered
When a customer enters contact information into a flow screen and clicks the Next button. When an employee clicks the Request PTO button on their employee record, which opens a form to complete. When a user clicks the Complete Sale button after closing an opportunity, and you want to start background automations, such as updating records and emailing stakeholders. When a user clicks a button on a form. When a user clicks a quick action button. When a user clicks a custom button or link.	Screen Screen Autolaunched
When a flow executes another flow within the same running instance to reduce repetition within the main flow. When an Apex class is triggered by a change to an opportunity's stage, which triggers an autolaunched flow. When an integrated printer is out of ink, and it publishes a platform event message. When a contact record is updated in NetSuite, and then the associated contact record in the connected Salesforce org is updated. When called by another flow. When called by Apex code. When a platform event message is received. When a change in the external system is found by a scheduled poll of the flow.	Autolaunched or Screen Autolaunched Platform Event-Triggered External System Change-Triggered
When a file is attached to an Account record. When an automation event such as a form submission is received. Automation Event-Triggered Flow	
Read at	Automate ur business process

Design the solution

1. Send Email Trigger: Contact created Action: Send welcome email	Challenge: Our sales reps forget to email new customers to welcome them. ●Solution: Send emails to new contacts automatically. ●Trigger: After a contact is created. ●Action: Send a welcome email to the new contact.
2. Create Records via Form Trigger: Agent submits screen form Action: Create Opportunity	Challenge: On customer calls, agents scramble to capture customer interest in new products. They often use sticky notes, which are hard to read or get lost. ●Solution: Use an online form embedded on the case record page to quickly capture basic information and create a new opportunity. The agent can confirm the information with the customer before ending the call. ●Trigger: When the agent submits the online form. ●Action: Use an online form to gather information, and then create an opportunity record when the agent submits the form.
Scheduled Membership Reminders Trigger: Daily schedule Action: Email customer + update membership status	Challenge: We can't keep track of our customers' memberships. Customers aren't notified before their memberships expire, and we lose out on renewals. ●Solution: Remind customers to renew their membership starting a month before it expires. ●Trigger: Check membership records each day to see if any are about to expire. ●Action: Email the customer when their membership is about to expire. Update the membership stage to pending so that you know which memberships are pending renewal.
External System → Salesforce Automation Trigger: NetSuite record created Action: Auto-create Lead in Salesforce	Challenge: Our sales reps currently create records in NetSuite, an external system, leading to a manual and often delayed process for integrating this new data into Salesforce. ●Solution: Implement an External System-Change Triggered Flow that automatically creates a corresponding record in Salesforce whenever a new record is created in the external system. ●Trigger: A new Contact record is created in NetSuite. ●Action: A new Lead record is automatically created in Salesforce, populated with relevant information from the NetSuite Contact
Salesforce → External System Automation Trigger: Salesforce Order created Action: Create Sales Order in NetSuite	Challenge: Our fulfillment center uses NetSuite, an external system, to process sales orders, while our sales team tracks them in Salesforce. The manual entry of sales orders into NetSuite from Salesforce leads to discrepancies, missing orders, and incomplete data. ●Solution: Implement a Salesforce Flow with an external action that automatically creates a new Sales Order in NetSuite whenever a new customer order is created in Salesforce.

	●Trigger: A new customer order is created in Salesforce. ●Action: A new Sales Order record is automatically created in NetSuite, populated with relevant information from the Salesforce Order, ensuring the fulfillment center receives accurate and timely data to complete the order.
Extract Legacy System Data Trigger: Button click Action: MuleSoft RPA extracts data and creates records	Challenge: Customer data exists in a legacy desktop application without API access. Employees manually copy and paste information into Salesforce, which is time-consuming and error-prone. ●Solution: Use MuleSoft RPA to extract data from the legacy application, and then create Salesforce records from your flow. ●Trigger: When a user initiates the data extraction process by clicking a button on a record page. ●Action: Call a MuleSoft RPA process that opens the legacy application, navigates to the customer data, extracts the information, and then creates the corresponding Salesforce records from within your flow.

IF YOU WANT TO:	FOR EXAMPLE:	USING THIS TOOL:	
Approve records through multiple levels in your organization.	Two levels of management to review and approve employee PTO request records.	Flow Approval Processes	
Suggest offers and actions to users that are tailored to meet your unique business criteria.	To prompt agents to offer service contracts to customers who don't have them.	Einstein Next Best Action	
Coordinate multiple flows, and assign them to multiple teams or individuals.	An automated hiring process that involves HR, Finance, and the hiring team.	Flow Orchestration	
Perform an operation for more records than scheduled-triggered flows allow.	To update all your 600,000 contacts at once.	Batch Apex	
Perform CPU-intensive operations.	To calculate a highly complex discount rate.	Apex	

Creating an Approval process

Approval processes	
A listing of both active and inactive approval processes for warranties is displayed below. To create a new approval process, click Create New Approval Process then select Use Jump Start Wizard to set up your approval process in a few short steps. Or, select Use Standard Wizard to configure all approval options.	 A listing of both active and inactive approval processes for warranties is displayed below. To create a new approval process, click Create process in a few short steps. Or, select Use Standard Wizard to configure all approval options.
	In the same way as workflow actions, approval actions consist of e-mail alerts, tasks, field updates, and outbound messages that can be triggered by the approval process.
Configure approval process	Providing the name of the process Specifying the entry criteria for the records Specifying who is going to approve Specifying the e-mail template Determining the fields to be displayed on the approver page Specifying who is going to send the approval mail
	1. Determine the steps and how many levels your process has. It is often useful to map out the process using a charting tool such as Microsoft Visio. 2. Decide if users can approve requests by e-mail and set up this feature accordingly. 3. Create an approval-request e-mail template. 4. Determine the approval-request sender. 5. Determine the assigned approver. 6. Determine the delegated approver, if necessary. 7. Decide if your approval process needs a filter. 8. Design initial submission actions. 9. Determine if users can edit records that are awaiting approval. 10. Decide if records should be auto-approved or rejected. 11. Determine the actions when an approval request is approved or rejected.

Approval process	An Approval Process in Salesforce is an automated workflow that routes a record to one or more people for approval.
	• Discounts need approval • PTO requests need a manager signature • Expenses must be reviewed • Contracts need multi-level approval
Entry criteria	Rules that decide which records are allowed to enter the approval process. Example: discount_c > 20%.
Initial submission actions	Record locking email alert field update Creating tasks
Record locking	Only admin or designated users can edit. This prevent users from modifying data mid approval.
Approval steps	Types of approval routing Submitter's manager. Queue specific user Related user (like account owner)

Action for each step	Update field(acc, rej), email, unlock record.
Approval submitter can recall.	I send the wrong document.
Reassign approval request	
Delegate approval	Approver can assign some1 to approve on their behalf.
Jump wizard	Great for: • Beginners • Simple approvals • One-level approvals • Quick setup 🔧 What it allows you to configure: • Entry criteria • Approver (single rule) • What actions happen on approval & rejection • Email notifications • Final approval actions ⚠️ What it does <i>not</i> allow: • Multiple approval steps • Complex routing logic • Parallel approvals • Step-level criteria So "Jump Start = Fast but Limited"
Standard wizard	Great for: • Multi-level approvals (Manager → Director → CFO) • Dynamic routing • Complex logic • Email alerts per step • Field updates per step • Parallel approvals (two people at same time) "Standard Wizard = Full Power Mode"

Creating an approval process

Manage Approval Processes For: Warranty_c A listing of both active and inactive approval processes for warranties is displayed below. To create a new approval process, click Create process in a few short steps. Or, select Use Standard Wizard to configure all approval options. Create New Approval Process	Create a new approval process. Mike selected jump start wizard.
Object manager > Warranty_c > fields(status)	Replacement Requested Replacement Accepted Replacement Rejected I added the following field
	Customer calls wanting to file a warranty claim because of a defective asset. We would have status designation at that point of replacement requested.
	The end user inside of salesforce can submit that for approval and then depending on whomever is the approver for that warranty claim can either select to accept that and approve that request and therefore then it will change to replacement accepted.
Approval Process Information The Jump Start wizard creates a one-step approval process for you in just a few minutes. Enter a name for your process in the box below and then select an email template to notify the approver (optional). Name Return Request Unique Name Return_Request Approval Assignment Email Template Add the Submit for Approval button and Approval History related list to all Warranty_c page layouts ☒	
Select Approver Operator 5 Using the options below, specify the user to whom the approval request should be assigned. ☒ Let the submitter choose the approver manually. ☐ Automatically assign an approver using a standard or custom hierarchy field: --None-- ☐ Automatically assign to queue. ☐ Automatically assign to approver(s). Because this is the Jump Start Wizard, Salesforce automatically chooses some settings for you. Show More	

Editing our Approval process

- 1

Create additional steps if a record requires more than one level of approval.
Example:
 - Discounts of more than 15% require an additional manager's approval
- 2

Add Initial Submission workflow actions. These actions will take place when a record is submitted for approval.
Example:
 - A Field Update action that changes the value of the Status field to "Pending"
- 3

Add Final Approval workflow actions. These actions will take place when a record has received all necessary approvals.
Examples:
 - A Field Update action that changes the value of the Status field to "Approved"
 - An Email Alert action that notifies the owner their record has been approved
- 4

Add Final Rejection workflow actions. These actions will take place when a record has been completely rejected by an approver.
Examples:
 - A Field Update action that changes the value of the Status field to "Rejected"
 - An Email Alert action that notifies the owner their record has been rejected
- 5

Add Recall workflow actions. These actions will take place when a submitted approval request is recalled.
Examples:
 - A Field Update action that changes the value of the Status field from "In Progress" to "Not Submitted"

Specify Entry Criteria

Use this approval process if the following criteria are met:

Field	Operator	Value
Warranty_c: Status	equals	Replacement Requeste

Entry level criteria

Select Field Used for Automated Approval Routing

Next Automated Approver Determined By: Manager

Use Approver Field of Warranty_c Owner: ☐

Record Editability Properties

☒ Administrators ONLY can edit records during the approval process.

☐ Administrators OR the currently assigned approver can edit records during the approval process.

Step 3. Specify Approver Field and Record Editability Properties

When you define approval steps, you can assign approval requests to different users. One of your options is to use a user field to automatically route these requests. If you want to use this option for any of your approval steps, select a field from the picklist below. Also, when a record is in the approval process, it will always be locked-- only an administrator will be able to edit it. However, you may choose to also allow the currently assigned approver to edit the record.

Available Fields

Asset

Created By

Currency

Description

End Date

Last Modified By

Start date

Status

warranty_provider

Add

Remove

Remove

Selected Fields

warranty Name

Owner

Up

Down

Step 5. Select Fields to Display on Approval Page Layout

Approval Page Fields

☒ Display approval history information in addition to the fields selected above.

Security Settings

☒ Allow approvers to access the approval page only from within the Salesforce application. (Recommended)

☐ Allow approvers to access the approval page from within the Salesforce application, or externally from a wireless-enabled mobile device.

	Step 6. Specify Initial Submitters Using the options below, specify which users are allowed to submit the initial request for approval. For example, expense reports should normally be submitted for approval only by their owners.
The owner of the approver is the one who can submit for approval. In submission settings, u can also recall from the queue.	Initial Submitters Submitter Type Search: Owner for: Find Available Submitters --None-- Allowed Submitters Warranty_c Owner Add Remove Submission Settings ☐ Allow submitters to recall approval requests

Activating our approval process

Select Activate to activate the approval process.	Process Definition Detail Edit Clone Deactivate Process Name Return Request Active ✓ Unique Name Return_Request Next Automated Approver Determined By Manager of Record Submitter Description Recently, 01 Station EQUALS Replacement Requested Entry Criteria Administrator OR Current Approver Allow Submitters to Recall Approval Requests ✓ Record Editability Support Case Created (Web Inquiries) Approval Assignment Email Template Warranty_c Owner Initial Submitters Created By Kazim Syed, 07/12/2025, 2:15 am Modified By Kazim Syed, 07/12/2025, 2:34 am
--	---

Return Request
Final Approval Record Lock

Save

Cancel

Choose Final Approval Record Locking Behavior

☐

 Lock the record from being edited

☒

 Unlock the record for editing

Save

Cancel

Because we want to allow user to make edits on status like replacement req to accepted or rejected.

Related **Details**

warranty Name

K-000001

C

Description

C

Asset

[Monitor](#)

II

warranty_provider

[Acme](#)

Start date

20/08/2025

End Date

29/08/2025

Status

Replacement Requested

Created By

[Kazim Syed](#), 30/08/2025, 1:46 am

L

We change the status of warranty record to “replacement req”.

Warranty Managem...

Home

warranties

Accounts

More

Submit for Approval

Warranty_c

K-000001

New Contact

Edit

Delete

View All

Approval History (0)

Activity

Clone

Change Owner

Printable View

Submit for Approval

Edit Labels

Warranty Managem...

Home

warranties

Accounts

Contacts

Contract Corner

Warranty_c

K-000001

New Contact

Edit

Delete

K-000001

[Kazim Syed](#)

Description

Currency

INR - Indian Rupee

Asset

[Monitor](#)

warranty_provider

[Acme](#)

Start date

20/08/2025

End Date

29/08/2025

Status

Replacement Requested

Created By

[Kazim Syed](#), 30/08/2025, 1:46 am

Last Modified By

[Kazim Syed](#), 07/12/2025, 2:41 am

Filters: All time · All act

Refresh

Submit for Approval

Upcoming & Overdue

No activities to show. Get started by sending an email, scheduling a task, and more.

No past activity. Past meetings and tasks marked as done show up here.

I Logged in as jim lindsay

Approval Processes

Warranty_c: Return Request

« [Back to Approval Process List](#)

Process Definition Detail

Process Name	Return Request	Active	☒
Unique Name	Return_Request	Next Automated Approver Determined By	Manager of Record Submitter
Description			
Entry Criteria	Warranty_c: Status EQUALS Replacement Requested		
Record Editability	Administrator OR Current Approver	Allow Submitters to Recall Approval Requests	☒
Approval Assignment Email Template	Support: Case Created (Web Inquiries)		
Initial Submitters	Warranty_c Owner		
Created By	Kazim Syed, 07/12/2025, 2:15 am		Modified By Kazim Syed, 07/12/2025, 2:38 am

Warranty_c
K-000003

Related

Details

warranty Name
K-000003

Owner
Jim Lindsay

Description
Joy stick aint working

Asset
Nintendo Oled

warranty_provider
Acme

Start date
07/12/2025

End Date
16/01/2026

Status
Replacement Requested

Created By
Jim Lindsay, 07/12/2025, 11:05 am

Last Modified By
Jim Lindsay, 07/12/2025, 11:05 am

You have to be the initial submitter to submit for approval.
So I had to create a record as Jim lindsay.

Submit for Approval

* Choose Next Approver

Jim Lindsay

X

To remove the selected record, press Backspace or Del.

Cancel

Submit

Notifications

[Mark all as read](#)

X

Jim Lindsay is requesting approval for warranty_c

warranty Name: K-000003 - Owner: Jim Lindsay

7 minutes ago

New versions of the latest PRM Reports and Dashboards are available

21-Nov-2025, 11:03 pm

Kazim

e suggestions

Approval Request Warranty_c Approval Pending Approve Reject Reassign Submitter Jim Lindsay Date Submitted 07-Dec-2025 Actual Approver Kazim Syed Assigned To Kazim Syed Details Approval Details warranty Name K-000003 Owner Jim Lindsay Submitter Comments Jim Lindsay Approval Process Maybe deactivated. Approver may be inactive Approver may not be in initial Submitter role 07-Dec-2025, 11:06:40 am

warranties > K-000003

Approval History

3 items - Sorted by Is Pending - Updated a few seconds ago

Step Name	Date	Status	Assign...	Actual ...	Comments
1 Step 1	07/12/2025, 11:30 am	Approved	Kazim Syed	Kazim Syed	Okay, We will replace the defective equipment.
2 Step 1	07/12/2025, 11:15 am	Reassigned	Kazim Syed	Laura Garza	Jim is really Dexter Morgan.
3 Approval Request Submitted	07/12/2025, 11:06 am	Submitted	Jim Lindsay	Jim Lindsay	Approval Process Maybe deactivated.Approver m

Create a Custom Hierarchical Relationship

Object manager > User object	
Data Type None Selected Select one of the data types below. Auto Number A system-generated sequence number that uses a display format you define. T Formula A read-only field that derives its value from a formula expression you define. T Roll-Up Summary A read-only field that displays the sum, minimum, or maximum value of a field i Hierarchical Relationship Allows users to click a lookup icon and select another user from a popup list. Click new fields	
	Self lookup to associate another user.

Field Label Warranty Return Approv Field Name Warranty_Return_Appro Description Help Text Child Relationship Name wra Error: You must enter a value	
Setup> Users → edit Jim lindsay user record	Additional Information Warranty Return Approver User Kazim Syed
Setup > Warranty Return Approver	When you define approval steps, you can assign approval requests to different users in your approval steps, select a field from the list to determine the next approver. You can also allow the currently assigned approver to approve the request. Select Field Used for Automated Approval Next Automated Approver Determined By Use Approver Field of Warranty_c Owner --None-- Standard User Fields ✓ Manager Custom User Fields Warranty Return Approver

Process Definition Detail

Edit

Clone

Deactivate

Process Name

Return Request

Active

✓

Unique Name

Return_Request

Next Automated Approver Determined By

Warranty Return Approver of Record Submitter

Description

Entry Criteria

Warranty_c: Status EQUALS Replacement Requested

Record Editability

Administrator OR Current Approver

Allow Submitters to Recall Approval Requests

✓

Approval Assignment Email Template

Support: Case Created (Web Inquiries)

Initial Submitters

Warranty_c Owner

Created By

Kazim Syed, 07/12/2025, 2:15 am

Modified By

Kazim Syed, 07/12/2025, 11:52 am

Approval Steps

Action	Step Number	Name	Description	Criteria	Assigned Approver	Reject Behavior
Show Actions Edit	1	Step 1			Manually Chosen	Final Rejection

Now instead of the Initial submitter to choose the assigned approver.

Select Approver

☐ Let the submitter choose the approver manually.

☒ Automatically assign using the user field selected earlier. (**Warranty Return Approver**)

☐ Automatically assign to queue.

☐ Automatically assign to approver(s).

☐ The approver's delegate may also approve this request.

	This document outlines the complete sequence of events, known as the Order of Execution , that Salesforce follows when a record is saved via an insert, update, or upsert DML statement (either through the user interface or API).
--	--

Salesforce Apex Trigger and Automation Order of Execution

This sequence ensures data integrity by enforcing validation, executing automation, and committing changes.

Step 1: Initialization and System Validation

1. **Load Record:** The original record is loaded from the database, or the record is initialized for an upsert.
2. **Overwrite Values:** The new record values provided by the user or API overwrite the old values.
3. **Client-Side Validation:** The browser runs JavaScript validation (for standard UI requests) for dependent picklist fields.
4. **Initial System Validation:** If the request is from a standard UI edit page, Salesforce runs basic system validations, including:
 - Compliance with layout-specific rules.
 - Required values and valid field formats.
 - Maximum field length.

Step 2: Before Automation and Validation (Before-Save)

5. **Before-Save Flows:** Executes **record-triggered flows** configured to run **before** the record is saved (Fast Field Updates).
6. **Before Triggers:** Executes all **before Apex triggers**.
7. **System & Custom Validation:** Runs most system validation steps again (verifying required fields, etc.), and runs any **custom validation rules**.
8. **Duplicate Rules:** Executes **duplicate rules**. If a duplicate is identified and the action is set to **Block**, the record is not saved, and the transaction stops.
9. **Foreign Keys:** Salesforce verifies that any custom

foreign keys do not refer to the object itself.

Step 3: Database Save (Uncommitted)

10.**Save Record:** The record is **saved to the database**, but the changes are **not yet committed** (they can still be rolled back).

Step 4: After Automation (After-Save)

11.**After Triggers:** Executes all **after Apex triggers**. (Used for updating related records or other complex logic).

12.**Assignment Rules:** Executes **assignment rules** (e.g., Lead or Case assignment).

13.**Auto-Response Rules:** Executes **auto-response rules**.

14.**Workflow Rules:** Executes **workflow rules**.

Step 5: Workflow Field Update Recursion (Re-evaluation)

15.**Workflow Field Updates:** If any Workflow Rules triggered a **field update** (a DML operation):

- The record is **updated again**.
- **System validations** are run again (Custom validation rules, Flows, Duplicate Rules, and Processes are **NOT** run again).
- **Before Update Triggers** and **After Update Triggers** are executed **one more time** (and only one more time).

Step 6: Processes and Other Rules

16.**Escalation Rules:** Executes **escalation rules**.

17.**Processes and Flows:** Executes the following (not in a guaranteed order):

- **Processes** built with Process Builder.
- **Flows** launched by processes or by workflow rules.
- **Record-Triggered Flows** configured to run **after** the record is saved (Actions and Related Records).

18.**Entitlement Rules:** Executes **entitlement rules**.

Step 7: Final Calculations and Commitment

19.**Roll-up Summary/Cross-Object Formulas:** If the

record contains a roll-up summary field or is part of a cross-object workflow:

- Calculations are performed, and the roll-up summary field in the **parent record** is updated.
- The **parent record** then goes through its own complete save procedure (starting from Step 1).

20.**Sharing Evaluation:** Executes **Criteria Based Sharing** evaluation.

21.**Commit to Database:** All DML operations are **committed** to the database.

Step 8: Post-Commit Logic

22.**Post-Commit Logic:** Executes final post-commit processes (asynchronous operations):

- Asynchronous Apex jobs (@future, Queueable Apex).
- Asynchronous paths in record-triggered flows.
- Sending emails.
- Outbound messages placed on the queue.

Important Considerations

- **Trigger Order:** If more than one trigger is defined on an object for the same event (e.g., two before insert triggers), **the execution order is not guaranteed**. Use a Trigger Framework to control execution.
- **API vs. UI:** The specific system validations that run may vary based on whether the request originated from a standard UI page or an API call.
- **Governor Limits:** All steps within one transaction (from the initial trigger to the final commit) are governed by the same set of **Apex Governor Limits** (e.g., 150 DML statements, 100 SOQL queries).

Question 1 – Workflow Automation

What are the key concepts of workflow automation in Salesforce CRM?

Select one: a) Workflow rules and time-dependent features.

b) Workflow rules and actions.

c) Workflow e-mail alerts and task actions.

d) Workflow field updates and task actions.

Question 2 – Workflow Processes

Which of the following processes can workflow rules be used to automate? Select all that apply:

a) Sending an e-mail alert to an account owner whenever updates are made to the account by another user.

b) Assigning a follow-up task to a marketing executive whenever a new lead is entered in the system.

c) Assigning a new lead to a queue.

d) Automatically updating an opportunity field whenever the opportunity reaches a specified sales stage.

Question 3 – Time-dependent Workflow

Which scenarios or criteria settings prevent you from adding time-dependent workflow actions? Select all that apply:

a) The rule is already active.

b) The rule criteria are set to evaluate when the record is created, and any time it's edited to

subsequently meet criteria.

c) The rule is deactivated, but has pending actions in the workflow queue.

d) The rule criteria are set to evaluate when the record is created, and every time it's edited.

Question 4 – Initiating Approval Processes

Which statements are correct when initiating an approval process?

Select all that apply:

a) Approval processes can be set to automatically filter which users are initially involved with the approval process.

b) Approval processes can be set to allow users to choose a user to send the approval to.

c) Approval processes can be set to allow users to choose a queue to send the approval to.

d) Approval processes show an error message if the record does not

	meet the entry criteria. e) Approval processes show a warning message showing the entry criteria if the record does not meet the entry criteria. Question 5 – Approving or Rejectioning approval processes Which statements are correct when approving or rejecting approval requests? Select all that apply: a) If enabled, approvers can approve or reject approval requests by e-mail. b) Approval requests must always be approved or rejected by a specified user. c) If enabled, approvers can approve or reject approval requests by Chatter. d) If enabled and a user is set, an approver's delegate can approve or reject approval requests.
	1) The answer is, a) Workflow rules and actions.
	Answer 2 – Workflow Processes The answer is, a) Sending an e-mail alert to an account owner whenever updates are made to the account by another user, b) Assigning a follow-up task to a marketing executive whenever a new lead is entered in the system, and ,d) Automatically update an opportunity field whenever the opportunity reaches a specified sales stage. The answer c) Assigning a new lead to a queue, is not possible using workflow rules and instead can be carried out using lead assignment rules (see Chapter 8, Introducing Sales Cloud, Service Cloud, and Collaborative Features of Salesforce CRM).
	Answer 3 – Time-dependent Workflow The answer is, a) The rule is already active (here, you must deactivate the rule first), c) The rule is deactivated, but has pending actions in the workflow queue (here, you must allow the pending actions to complete or delete the pending action using the time-based workflow queue, and, d) The rule criteria is set to evaluate when the record is created, and every time it's edited.
	Answer 4 – Initiating Approval Processes

	The answer is, a) Approval processes can be set to automatically filter which users are initially involved with the approval process, b) Approval processes can be set to allow users to choose a user to send the approval to, and d) Approval processes show an error message if the record does not meet the entry criteria. The answer, c) and e) are not correct for the following reasons, respectively: Approval processes can be set to automatically assign approvers from a queue but not to allow users to choose a queue to send the approval to. Approval processes show an error message only, and not a warning message showing the entry criteria if the record does not meet the entry criteria.
	Answer 5 – Approving or Rejecting approval Processes The answer is, a) If enabled, approvers can approve or reject approval requests by e-mail, c) If enabled, approvers can approve or reject approval requests by Chatter, and d) If enabled and a user set, an approver's delegate can approve or reject approval requests. The answer b) is not correct because approval requests can be set to be approved or rejected by individuals in a queue.
	Answer 6 – During Approval Processes The answer is, c) During an approval process approvers can reassign approval requests and d). During an approval process, approvers can view approval requests on their Salesforce homepage.

Perfect — let's go **slow, deep, and exam-oriented**.

We'll tackle your confusion on **Q1, Q6, Q10, Q11, Q20**, explain them in a way that *builds intuition*, and then give you **new practice questions of similar difficulty**, each with clear explanations.

FIRST — Clarifying the 5 Confusing Questions

1) Why is Q1 = BEFORE SAVE?

Question Recap:

A Flow needs to update a field on the *same record* that triggered it. Which is most efficient?

Correct answer: **Before-save Record-Triggered Flow**

Why?

Because **before-save flows can update the same record very efficiently** *without performing DML*.

Important rule:

Updating the triggering record → use BEFORE-SAVE

Updating related records → use AFTER-SAVE

Why your reasoning ("it won't trigger again") is incorrect?

"Before save won't trigger itself again" — true,
but that's not why we choose it.

We choose it because:

- It is **FASTEST automation in Salesforce**
- It performs **implicit DML**
- It is **designed** to modify the same record efficiently

This is an **ADM-201 gold rule**.

6) Q6 Explanation — After Save

Scenario:

When Account Industry changes, create tasks for ALL related Contacts.

Correct Answer: **After-Save Flow**

Why?

Because:

- You are updating *related records* (Contacts → Tasks)
- You cannot do that in **before save** (only the main record)
- You need the record to be committed to DB before creating related records

Rules:

Need to... Flow Type Update same record Before Save Create related records **After Save**

10) Q10 Explanation — Record-Triggered BEFORE Delete

Question:

Flow must run when a record is deleted.

Correct answer: **Before Delete**

Why not After Delete?

Because **Salesforce does NOT support After-Delete Flows** (ADM-201 fact).

Only available deletion trigger:

✓ **Before Delete**

11) Q11 Explanation — Schedule-Triggered Flow

Scenario:

You want to mass-update Accounts missing Industry values.

Correct answer: **Scheduled Flow**

“But isn’t that a one-time action? Why schedule?”

Because ADM-201 is testing **capability**, not intention.

You pick Scheduled Flow because:

- It processes **multiple records at once**
- It runs **in batch mode**
- It doesn’t need a user to make updates
- It is the **only flow type that can run across the entire database on a schedule**

Even if you want it only ONCE, the *correct tool* is still a **Schedule-Triggered Flow**.

20) Q20 Explanation — C is correct

Question Recap:

A flow runs when a user clicks a button.

Options:

- Screen Flow
- Autolaunched Flow
- Either
- None

Correct: **Either**

Why?

A button can trigger:

- A **Screen Flow** (opens a wizard)
- An **Autolaunched Flow** (runs silently)

Button is just a **trigger method**, not an interaction.

NOW — NEW QUESTIONS (similar difficulty)

Each question has **full reasoning**.

New Question Set (High Difficulty, ADM-201 Style)

Q1. A Flow must update the record AND create a Case when an Opportunity Stage changes to “Escalated”. Which Flow design is best?

- A. One before-save flow
- B. One after-save flow
- C. Two before-save flows
- D. One before-save + one after-save

Correct Answer: D

Explanation

- Before Save → update Opportunity
 - After Save → create Case (requires record commit)
- You can't do both in one flow.
-

Q2. A Flow must check if a Contact has more than 3 open Cases and prevent the user from saving the Contact. Which flow type?

- A. After Save Record-Triggered
- B. Before Save Record-Triggered
- C. Autolaunched Flow
- D. Screen Flow

Correct Answer: B

Why?

Before Save flows can:

- Validate the record
- Prevent save via error message

After Save cannot block a save.

Q3. You need automation that runs when a Platform Event arrives and updates 10,000 records. Which Flow type?

- A. Autolaunched
- B. Record-Triggered
- C. Platform Event-Triggered Flow
- D. Schedule-Triggered Flow

Correct Answer: C

Why?

Only Platform Event Flows respond to platform events.

Q4. A manager wants to run a flow manually from the Setup menu to clean up old Leads. Which flow type?

- A. Screen Flow
- B. Autolaunched Flow

- C. Record-Triggered Flow
- D. Schedule-Triggered Flow

Correct Answer: B

Why?

Autolaunched Flows can be:

- Manually run from the Flow Runtime UI
- Triggered on demand
- Run without screens

Screen Flow requires UI interaction.

Q5. A user clicks a button and is presented with a guided 3-step wizard. Which flow?

- A. Autolaunched
- B. Screen Flow
- C. Record triggered
- D. Scheduled flow

Correct Answer: B

Why?

Only Screen Flows display UI screens.

Q6. A flow needs to update 50,000 Contacts every night based on custom logic. How should you build it?

- A. Before-save record triggered
- B. After-save record triggered
- C. Autolaunched + manual
- D. Schedule-triggered flow

Correct Answer: D

Explanation

Scheduled flows support:

- Running daily
- Batch processing
- Large volumes
- Background processing

Q7. A flow must run *only* if a field changes from “Draft” to “Submitted”. Where do you configure this?

- A. Decision element
- B. Start element entry criteria
- C. Assignment
- D. Loop

Correct Answer: B

Why?

Record-triggered flow start conditions include:

- When the flow runs
- Field change conditions
- Old → new comparison logic

Q8. Which flow action stores temporary values that are used later in the flow?

- A. Elements
- B. Resources
- C. Fault Paths
- D. Subflows

Correct Answer: B

Why?

Resources = variables.

Q9. A Flow needs to interact with the user AND update fields. Which type?

- A. Before Save Flow
- B. Screen Flow
- C. Autolaunched Flow
- D. After Save Flow

Correct Answer: B

Q10. A Flow triggered by a record update must not run every time — only when a specific field value changes. What feature should you use?

- A. Loop
- B. Subflow
- C. Entry conditions
- D. Fault path

Correct Answer: C

If you'd like, I'll generate:

✓ 20 more (even harder) Flow scenario questions,

✓ A 1-page ADM-201 Flow Exam Cheat Sheet,

✓ or a Flow Decision Tree (which flow type to choose, instantly).

Just tell me which one you want!